Knuth Division

Time limit: 1.00 s **Memory limit:** 512 MB

Given an array of n numbers, your task is to divide it into n subarrays, each of which has a single element.

On each move, you may choose any subarray and split it into two subarrays. The cost of such a move is the sum of values in the chosen subarray.

What is the minimum total cost if you act optimally?

Input

The first input line has an integer n: the array size. The array elements are numbered $1, 2, \dots, n$.

The second line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print one integer: the minimum total cost.

Constraints

- $1 \le n \le 5000$
- $1 \le x_i \le 10^9$

Example

Input	Output
5 2 7 3 2 5	43